# Summer Math Adventure

W3
D1

Week 3 · Day 1 | Multi-Digit Multiplication: Mastery

Name: ______________________ Date: ______________________

MULTIPLICATION

**Standard Algorithm – 2-digit × 2-digit:**
47 × 38: Multiply 47 × 8 = 376, then 47 × 30 = 1,410. Add: 376 + 1,410 = 1,786.
**Shortcut check:** Estimate first! 47 × 38 ≈ 50 × 40 = 2,000. Answer 1,786 is close. ✓
**Area model alternative:** Split 47 = 40 + 7, 38 = 30 + 8. Fill in the 4 boxes and add.

## Part A – Standard Algorithm

1. 73 × 46 =

2. 258 × 37 =

3. 614 × 83 =

4. 307 × 59 =

5. 4,025 × 14 =

6. 528 × 406 =

## Part B – Area Model

*Fill in the area model, then add the parts.*

7. 63 × 47 using the area model:

| | 40 | 7 |
|---|---|---|
| 60 | | |
| 3 | | |

Total: ___ + ___ + ___ + ___ =

## Part C – Word Problems (Multi-step)

8. A theater has 48 rows of seats. Each row has 36 seats. Tickets cost $12 each. If the theater is completely full, how much money is collected?

Answer: $

9. A car travels 58 miles per hour. How far does it travel in 24 hours? (Cumulative: round to nearest hundred)

# Summer Math Adventure

W3 D2

Exact:

Week 3 · Day 2 | Long Division: 4-Digit ÷ 2-Digit

Name: ______ Date: ______

LONG DIVISION

**Long Division Steps — remember "DMSB":**

**D**ivide → **M**ultiply → **S**ubtract → **B**ring down (then repeat)

Example: 1,344 ÷ 16 → Ask: how many 16s in 13? → 0. How many 16s in 134? → 8 (8×16=128). Subtract → 6. Bring down 4 → 64. 64÷16=4. Answer: 84.

**Always check:** quotient × divisor + remainder = dividend.

Work (try different digits):

Missing digit: ☐

## Part A – Divide (show each step)

1. 5,376 ÷ 24 =

Check:

2. 7,056 ÷ 48 =

Check:

3. 4,891 ÷ 13 =

Q:

R:

4. 9,072 ÷ 36 =

Check:

## Part B – Estimate the Quotient First

*Round the divisor to a friendly number, estimate, then calculate exactly.*

5. 8,463 ÷ 27 Estimate:

Exact:

6. 6,480 ÷ 32 Estimate:

# Summer Math Adventure

Exact:

W3 D3

Week 3 · Day 3 | Factors, Multiples, Prime & Composite Numbers

Name: ____________ Date: ____________

**FACTORS & MULTIPLES**

**Factor:** a number that divides evenly into another. Factors of 12: 1, 2, 3, 4, 6, 12.
**Multiple:** the result of multiplying by a whole number. Multiples of 6: 6, 12, 18, 24…
**Prime:** exactly 2 factors (1 and itself). Examples: 2, 3, 5, 7, 11, 13…
**Composite:** more than 2 factors. Example: 12 has 6 factors → composite.
**GCF:** Greatest Common Factor. **LCM:** Least Common Multiple.

Answer:

## Part A – List All Factors

1. Factors of 36:

2. Factors of 48:

8. 5,040 students are divided equally into 14 classrooms. Each classroom has __ students. Then each group of 4 students shares one table. How many tables in one classroom?

3. Factors of 60:

4. Factors of 100:

Students per room:

## Part B – Prime or Composite?

*Write P (prime) or C (composite). For composite numbers, list one factor pair (other than 1 × itself).*

Tables per room:

| Number | P or C? | Factor pair (if composite) |
|---|---|---|
| 17 | | |
| 24 | | |
| 37 | | |
| 49 | | |
| 51 | | |
| 97 | | |

⭐ **Challenge – Division Riddle**

I am a 4-digit number. When divided by 25, my quotient is 148 and remainder is 0. When divided by 4, my remainder is 0. What am I?

Work:

Answer:

## Part C – GCF and LCM

*Use factor lists or prime factorization (factor trees).*

5. GCF of 24 and 36 = ☐

6. GCF of 45 and 60 = ☐

Factor trees or lists:

7. LCM of 4 and 6 = ☐

8. LCM of 8 and 12 = ☐



## Part D – Word Problems: GCF & LCM in Action

9. Ms. Rivera has 36 math books and 48 science books. She wants to make equal piles using all the books, with each pile having only one type of book. What is the greatest number of books per pile?

# Summer Math Adventure

W3 D4

Week 3 · Day 4 | Order of Operations (PEMDAS)

Answer:

Name: ____________________ Date: ____________________

ORDER OF OPERATIONS

books per pile, making

**PEMDAS:** **P**arentheses → **E**xponents → **M**ultiply & **D**ivide (left to right) → **A**dd & **S**ubtract (left to right)

Example: $3 + 4 \times (2 + 1)^2 - 5$

$= 3 + 4 \times 3^2 - 5$ (parentheses first)

$= 3 + 4 \times 9 - 5$ (exponents)

$= 3 + 36 - 5$ (multiply)

$= 34$ (left to right add/subtract)

at 0 seconds. When is the next time they blink together?

## Part A – Evaluate Step by Step

*Show EACH step. Circle the operation you do first.*

1. $5 + 3 \times 4 =$

Step 1:

Answer: after

____________________

seconds.

Step 2:

____________________

2. $(5 + 3) \times 4 =$

Step 1:

____________________

Step 2:

____________________

### ⭐ Challenge – The Sieve of Eratosthenes (mini)

Cross out all multiples of 2, 3, 5, and 7 in the grid below (don't cross out 2, 3, 5, 7 themselves). The remaining numbers are prime!

3. $(20 - 12) \div 4 + 2 =$

4. $(20 - 12) \div 4 + 2 =$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 |
| 47 | 48 | 49 | 50 | | | | | |

5. $3^2 + 4 \times (6 - 2) =$

6. $48 \div (2 \times 3) + 5^2 =$

List all primes you found between 1–50:

## Part B – Add Parentheses to Make It True

*Place parentheses in each expression to make the equation correct. You may need to try a few spots.*

7. $3 + 5 \times 2 - 1 = 15$

8. $10 - 2 \times 3 + 1 = 25$

9. $24 \div 4 + 2 \times 3 = 12$

10. $5 \times 3 + 4 - 2 = 33$

## Part C – Write Your Own



11. Write an expression using + , × , and parentheses that equals 36. Use at least 4 numbers.

# Summer Math Adventure

W3
D5

Week 3 · Day 5 | Multiplication & Division Word Problems + Full Review

Name: ______________________ Date: ______________________

**OPERATIONS REVIEW**

**Word problem strategy:** Read → Identify what's asked → Choose operation → Estimate → Solve → Check if answer makes sense.

## Part A – Mixed Computation Sprint

1. 784 × 56 =

2. 9,408 ÷ 28 =

3. $5 \times 3^2 - (4 + 2) =$

4. GCF(18, 30) = ______________________

5. LCM(6, 9) = ______________________

6. Is 91 prime? Circle: YES / NO
Factor pair if composite: ______________________

## Part B – Multi-Step Word Problems

7. A pool holds 35,280 gallons. A pump fills it at 840 gallons per hour. How many hours will it take to fill the pool completely?

Answer: ______________________ hours

8. Marcus earns $14 per hour. He works 38 hours one week and 42 hours the next. He spends $350 on rent. How much does he have left after paying rent over both weeks?

Answer: $ ______________________

9. A store has 2,400 apples, 1,800 oranges, and 3,000 bananas. They want to pack fruit bags where every bag has an equal number of each type of fruit. What is the greatest number of bags they can make? (Hint: use GCF)

## Part D – Cumulative Word Problem

12. At a bake sale, cupcakes cost $3 each and cookies cost $1 each. Javier buys 4 cupcak and 6 cookies, then shares the cost equally with a friend. Write an expression with parentheses and solve.

Each person pays: $ ______________________

### ⭐ Challenge – Make 24

Use the numbers **3, 3, 8, 8** (each exactly once) with **+**, **–**, **×**, **÷** and parentheses to make **24**.

Try different combinations:

Solution:

Answer:

bags, each with

apples,

oranges,

bananas.

# Week 3 Quiz – Multiplication, Division & Number Theory

W3 QUIZ

Topics: Multi-digit ×÷ · Factors/Multiples · GCF/LCM · Order of Operations

Name: ______________________ Date: ______________________

Score: _______ / 20

## Section 1 – Multiplication & Division (8 pts)

## Part C – Cumulative Review (Weeks 1–3)

1. 437 × 68 =

2. 6,720 ÷ 32 =

10. Round 4,709,283 to the nearest ten-thousand:

11. $\frac{7}{8} - \frac{1}{3}$ =

12. $3\frac{2}{5} + 1\frac{7}{10}$ =

13. $\frac{4}{9} \times \frac{3}{8}$ =

3. 5,036 ÷ 14 = Q: ___ R: ___

4. 4 × (3 + 7)² ÷ 5 =

### ⭐ Challenge – Puzzle Grid

Fill in the grid so every row and column multiplies to give the product shown.

| | ×? | ×? | Product |
|---|---|---|---|
| ×? | 4 | | 20 |
| ×? | | 6 | 18 |
| Product | 12 | 30 | |

## Section 2 – Factors, Primes, GCF, LCM (6 pts)

5. List all factors of 42:

6. Is 83 prime or composite?

7. GCF(32, 48) =

8. LCM(9, 12) =

Work:

9. **(2 pts)** You have 24 red beads and 36 blue beads. You want to make identical bracelets using all beads (each bracelet has the same mix). What is the maximum number of bracelets? How many of each color per bracelet?

Bracelets:

Red each:

Blue each:

## Section 3 – Word Problems (6 pts)



10. **(3 pts)** A farmer plants 45 rows of corn with 68 plants in each row. He harvests 2,856 plants and stores them in bags of 24 plants each. How many full bags does he fill?

Plants total:

______________________

Full bags:

______________________

11. **(3 pts)** Write and evaluate an expression: A store sells books for $8 and pens for $2. Julia buys 5 books and 7 pens, then gets $4 off. Write the expression and find the total.

Expression:

________________________________________

Total: $

______________________